# Charvi Patel

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## Education

**Royal Holloway, University of London** — Sep 2024 – Jun 2027
*Bachelor of Science in Computer Science — Average Grade: 72% (1st)* — *London, UK*

**St Dominic's Sixth Form College** — Sep 2022 – Jun 2024
*A Levels: Mathematics (B), Further Mathematics (B), Computer Science (C)* — *London, UK*

**Bentley Wood High School for Girls** — Sep 2017 – Jun 2022
*GCSEs: 11 A*–B including Mathematics and English Language* — *London, UK*

## Technical Projects

**Low-Latency Concurrent TCP Job Scheduler — C++20, Linux, TCP, epoll** *Sep 2026*
- Built a concurrent TCP job scheduler in C++20 on Linux using non-blocking sockets and epoll, benchmarking workloads of 10,000+ scheduled jobs
- Implemented a configurable worker thread pool and thread-safe priority queue using mutexes and condition variables to support concurrent job scheduling and execution
- Developed sequential and concurrent execution modes and benchmarked 10,000+ jobs across worker configurations, measuring throughput, average latency, and p95/p99 latency
- Built automated GoogleTest coverage for priority scheduling, concurrent queue access, worker execution, and graceful shutdown

**Neuro-Route — 1st iCHack Imperial Hackathon** *Feb 2026*
- Led a team of 4 to build a sensory-friendly journey planner for neurodivergent travellers within 48 hours
- Developed a Python/FastAPI backend integrating TfL, Bus Open Data Service, and DfT Statistics APIs to generate personalised journey options
- Implemented route scoring across travel time, crowding, noise, and heat, evaluating approximately 50 route combinations per journey
- Won 1st place for Best Travel Hack and 3rd place in IMChange: Tech for Good at iCHack Imperial

**Restaurant Ordering System — Java & Spring Boot** *Jul 2025 – Apr 2026*
- Developed a full-stack restaurant ordering system using Java and Spring Boot, designing REST APIs for ordering, menu management, staff workflows, and administration
- Designed role-based access control with Spring Security across customer, staff, and administrator permissions
- Applied Test-Driven Development with JUnit and automated GitLab CI pipelines, using Checkstyle and SpotBugs for code quality and static analysis

## Extracurricular Activities

**President — Women in STEM Society, Royal Holloway** *Sep 2026 – Present*
- Lead a society committee delivering networking events, technical workshops, and professional development activities engaging 100+ students
- Develop relationships with industry professionals and companies to create guest speaker, partnership, and career opportunities for members

**Citi Global Markets Trading Simulation — 1st Place** *Sep 2025*
- Grew a simulated $10M portfolio to $13.8M through equities and derivatives trading, achieving a 38% return in a competitive global markets simulation
- Analysed market movements and managed equity and derivatives positions to balance potential returns against risk under time pressure

## Additional Information

**Languages**: English (Native), Gujarati (Fluent), Hindi (Intermediate)
**Technical Skills**: C++, Python, Java, C#, TypeScript, JavaScript, SQL, Linux, TCP/IP, epoll, Spring Boot, React, Flask, Flutter, Docker, CMake, GoogleTest, JUnit, Git, GitHub, GitLab CI, Jira
**Interests**: Strength training, travelling, language learning